



Interactive MF Analytics Dashboard

Streamlit Web App • Day 5 Bonus Challenge • Bluestock Fintech Internship

Project Overview

This capstone project analyses the Indian Mutual Fund industry using real AMFI data, mfapi.in live NAV feeds, and a synthetic investor transaction dataset. The full pipeline - ETL, SQL database, EDA, performance metrics, and advanced analytics - is built in Python and exposed through this interactive Streamlit dashboard as an alternative to Power BI (Bonus Challenge, +10 marks).

Dashboard Structure

6 analytical pages with interactive slicers and CSV export:

1 Industry Overview

Total AUM 81L Cr, SIP 31,002 Cr, 26.12 Cr folios, 1,908 schemes. AUM trend 2022-2025 and top 10 fund houses.

2 Fund Performance

Risk-return scatter, composite scorecard, NAV vs benchmark. Slicers: Fund House, Category, Plan.

3 Investor Analytics

Transaction by state, SIP vs Lumpsum split, age-group SIP, monthly volume. Slicers: State, Age Group, City Tier.

4 SIP & Market Trends

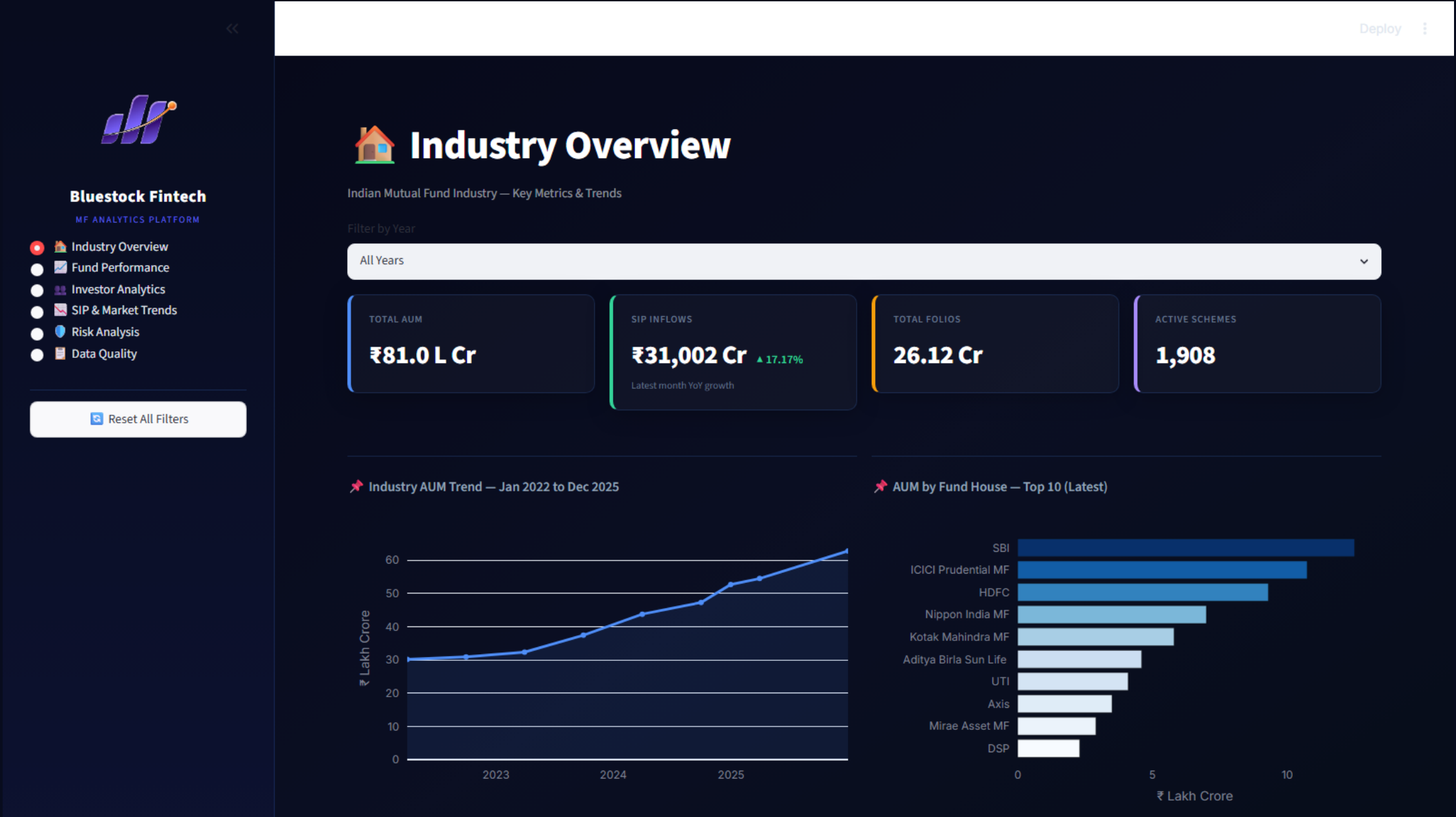
SIP inflow vs Nifty 50 dual-axis, category heatmap, top 5 categories FY25, YoY SIP growth KPI.

5 Risk Analysis

VaR 95% & CVaR, Max Drawdown ranking, Alpha/Beta table, rolling 90-day Sharpe. Slicers: Category, Risk Grade.

6 Data Quality

Dataset completeness, null-value counts, duplicate audit, AMFI code validation across all 10 raw datasets.





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MF ANALYTICS PLATFORM

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP & Market Trends
- Risk Analysis
- Data Quality

Reset All Filters



Fund Performance Analysis

Risk-Return Analysis, Scorecard & NAV vs Benchmark

Fund House

All

Category

All

Plan

All

TOP PERFORMER (3YR CAGR)

35.10%

Axis Midcap

HIGHEST SHARPE RATIO

1.46

Mirae Asset Large Cap

AVG ALPHA

+15.91%

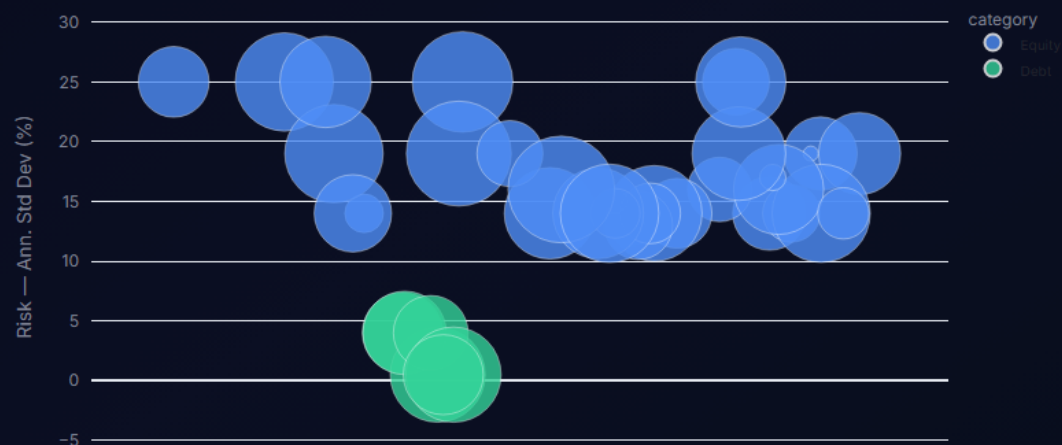
Excess return vs benchmark

POSITIVE ALPHA FUNDS

40/40

Outperforming benchmark

Scatter: Return (X) vs Risk / StdDev (Y) — bubble size = AUM



Top 8 Funds by Composite Score





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Investor Analytics

Transaction Patterns, Demographics & Geography

State

All

Age Group

All

City Tier

All

UNIQUE INVESTORS

5,000

TOTAL VOLUME

₹352.16 Cr

AVG SIP AMOUNT

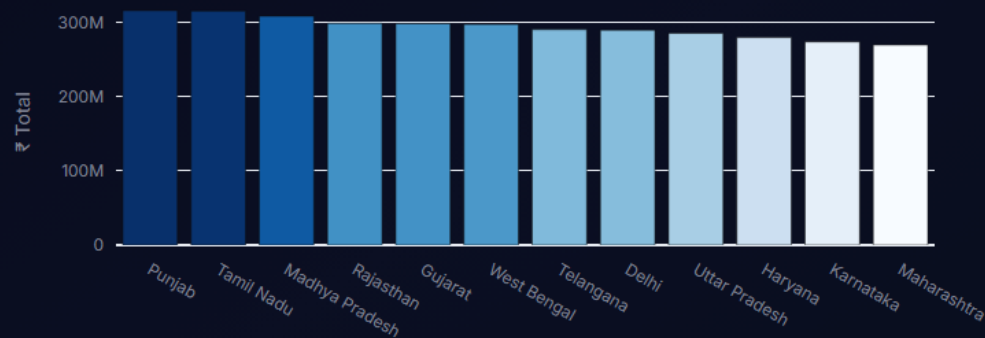
₹11,018

T30 CITY SHARE

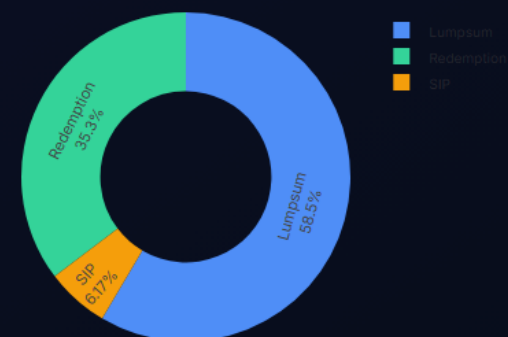
65.9%

B30: 34.1%

Transaction Amount by State (Top 12)



SIP vs Lumpsum vs Redemption



Avg SIP Amount by Age Group

Deploy





Deploy



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SIP & Market Trends

SIP Growth, Category Flows & Benchmark Correlation

Year

All Years

Category

All Categories

LATEST SIP INFLOW

₹31,002 Cr ▲ 17.17%

ACTIVE SIP ACCOUNTS

9.35 Cr

Latest month

PEAK SIP MONTH

2025-12

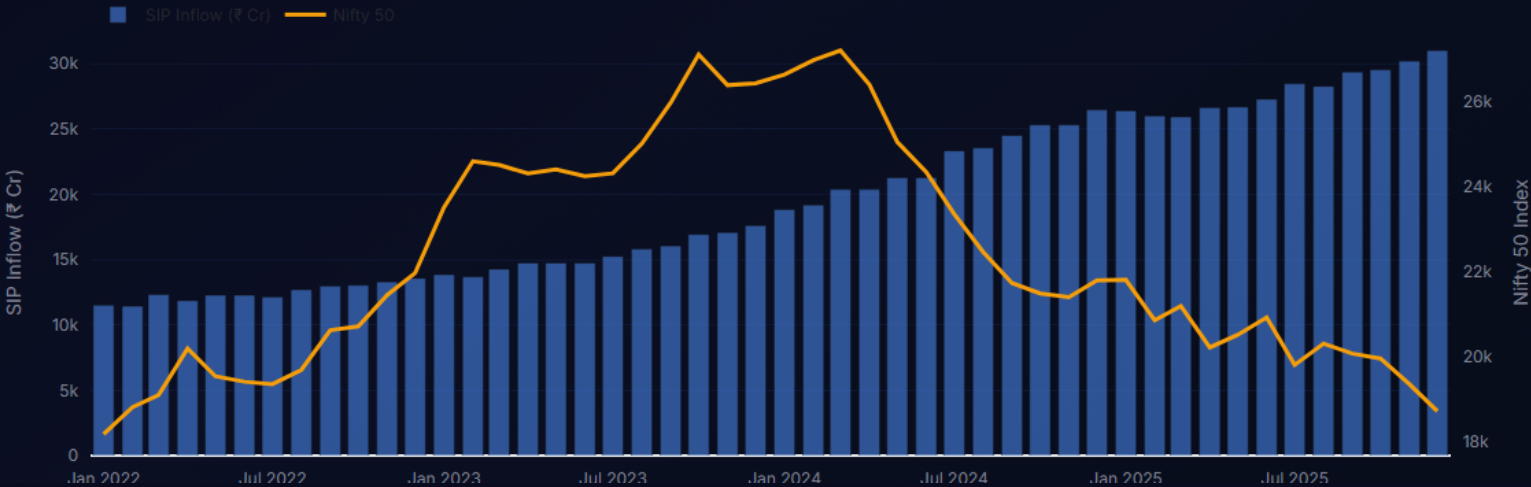
₹31,002 Cr

MOST POPULAR CATEGORY

Liquid

By total net inflow

Dual-Axis: SIP Inflow (Bar) + Nifty 50 Index (Line) — 2022 to 2025





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 Reset All Filters

VaR, CVaR, Drawdowns, Sharpe and Benchmark Correlation Matrix

All

AVG TRACKING ERROR

20.36%

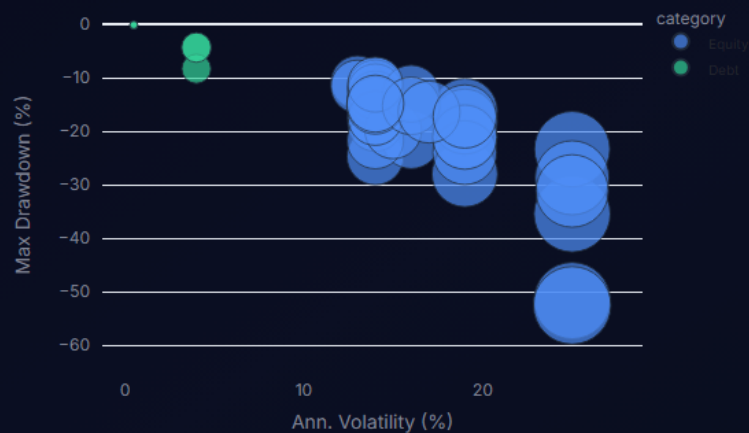
MEDIAN 95% VAR

1.44%

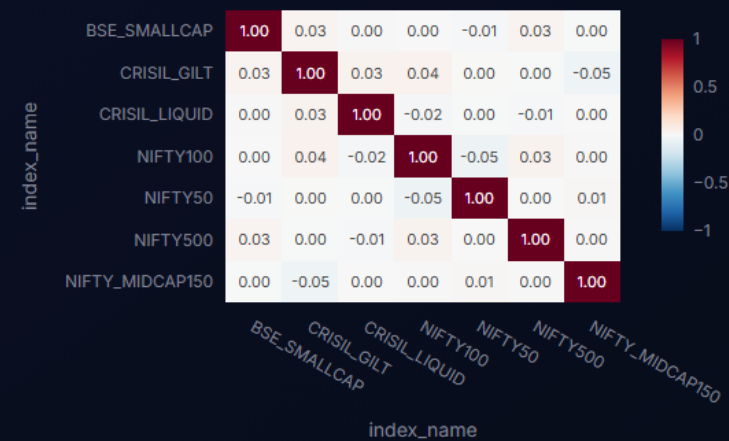
WORST MAX DRAWDOWN

-52.57%

📌 Volatility vs Max Drawdown (bubble = VaR)



📌 **Benchmark Correlation Matrix (Daily Returns)**



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MF ANALYTICS PLATFORM

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Data Quality

Reset All Filters

Data Quality Monitor

Schema Conformity, Completeness & Freshness Dashboard

TABLES LOADED

10

TOTAL RECORDS

87,533

DUPLICATE ROWS

0

NULL VALUES

12

Quality Matrix

Table	Rows	Cols	Duplicates	Nulls	Date Range	Status
01_fund_master	40	15	0	0	1996-09-11 → 2015-12-28	OK
02_nav_history	46000	3	0	0	2022-01-03 00:00:00 → 2026-05-29 00:00:00	OK
03_aum_by_fund_house	90	5	0	0	2022-03-31 00:00:00 → 2025-12-31 00:00:00	OK
04_monthly_sip_inflows	48	6	0	12	2022-01 → 2025-12	Warning
05_category_inflows	144	3	0	0	2024-04 → 2025-03	OK
06_industry_folio_count	21	6	0	0	2022-01 → 2025-12	OK
07_scheme_performance	40	19	0	0	-	OK
08_investor_transactions	32778	14	0	0	2024-01-01 00:00:00 → 2025-05-30 00:00:00	OK
09_portfolio_holdings	322	8	0	0	2025-12-31 → 2025-12-31	OK
10_benchmark_indices	8050	3	0	0	2022-01-03 00:00:00 → 2026-05-29 00:00:00	OK

Download CSV